The Prime Axis as a Circle: Fermat's Little Theorem via the Multiplicative Group of $\mathbb{Z}/p\mathbb{Z}$ — a Pure-Known-Mathematics Re-formalization

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Abstract

We report the second completed exercise of a re-formalization program ("0pat"): a pat framework insight ("the prime axis, periodized, is a circle; under the Euler-circle view the prime axis itself is a circle") is re-stated in pure known mathematics and formally proved in Lean 4 / mathlib with *no* pat framework concepts and *no* complex plane. The re-statement is the algebra of the multiplicative circle group $(\mathbb{Z}/p\mathbb{Z})^\times$: the circle's size ($p-1$ elements), the circle's algebra (Fermat's little theorem, both the coprime form $a^{p-1} \equiv 1$ and the complete form $a^p \equiv a \pmod p$), and the global structure (infinitely many primes). The proof is four theorems built entirely on mathlib: an explicit bijection $\{x \neq 0\} \simeq \mathrm{Fin}(p-1)$, Lagrange's theorem via `orderOf_dvd_card` and `pow_orderOf_eq_one`, the congruence-lifting `Nat.ModEq.pow`, and `Nat.exists_infinite_primes`. All four compile with zero errors and zero `sorry` against mathlib v4.32.2.

Keywords: Lean, formalization, Fermat's little theorem, modular arithmetic, circle group, 0pat, reproducibility

1 Introduction

Program. This paper is the second completed exercise of a re-formalization program ("0pat"): take the exercises of the pat framework — a body of Lean-formalized observations about basepoint- relative operator semantics — and re-prove their mathematical content using *pure known mathematics*: only mathlib's existing definitions and theorems, no pat framework concepts (basepoints, phase locking, interlock pairs, projection superpositions).

Method. Each exercise is located on a genealogy map (rows: pat concepts R1–R25; columns: mathematical genealogies). The map's entries are the paths: a pat concept at coordinate (R_i, C_j) has a standard-mathematics correspondent at that coordinate. The re-formalization walks the path: re-state the pat claim at its coordinate in standard mathematics, formalize the equivalence, and prove.

This exercise. The pat insight under study arose in the Riemann-direction work: the number line, periodized modulo p, curls into the circle $\mathbb{Z}/p\mathbb{Z}$; in this "Euler-circle" view the prime axis itself is a circle, and its structure should be observable by the same method used for the Riemann direction — without needing the complex plane. The insight sits at $(R23, C3)$ on the genealogy map (prime circle × analytic number theory). Its standard-mathematics correspondent is the algebra of the multiplicative circle group $(\mathbb{Z}/p\mathbb{Z})^\times$: Fermat's little theorem and the infinitude of primes. We re-state and prove it.

2 Re-statement (0pat)

Pat claim. The prime axis, periodized, is a circle; on the circle, the prime structure is the multiplicative circle group: every nonzero residue is invertible (the circle is a group of size $p-1$), and the circle's algebra is $x^p = x$ (Fermat's little theorem). The prime axis is not exhausted by any finite modulus: there are infinitely many primes.

0pat re-statement. Let p be prime.

1. *Circle size:* $|(\mathbb{Z}/p\mathbb{Z})^\times| = p - 1$ (the field $\mathbb{Z}/p\mathbb{Z}$ is a domain, so every nonzero element is a unit).
2. *Circle algebra (coprime form):* if $\gcd(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod p$ — the exponent of the circle group divides the group's order (Lagrange).
3. *Circle algebra (complete form):* for *every* a, $a^p \equiv a \pmod p$ — including $p \mid a$, where both sides vanish modulo p.
4. *Global structure:* for every n, there is a prime $p \geq n$ (the prime axis is infinite).

3 Formalization

The proof is a single Lean file (`proof.lean`, 212 lines), importing `Mathlib.Data.ZMod.Basic`, `Mathlib.GroupTheory.OrderOfEl` and `Mathlib.Data.Nat.Prime.Basic`. Four theorems:

Theorem	Statement	math
`card_units_zmod_prime`	$\lvert(\mathbb{Z}/p\mathbb{Z})^\times\rvert = p - 1$	expl
`fermat_little`	$\gcd(a, p) = 1 \Rightarrow a^{p-1} \equiv 1 \pmod p$	`orde`
`fermat_little_complete`	$\forall a,\ a^p \equiv a \pmod p$	`Nat.`
`infinitely_many_primes`	$\forall n,\ \exists p \geq n$, prime p	`Nat.`

Circle size by explicit bijection. Instead of appealing to mathlib's unit-coprime equivalence, the proof constructs the bijection $\{x : \mathbb{Z}/p\mathbb{Z} \mid x \neq 0\} \simeq \mathrm{Fin}(p-1)$ directly: $x \mapsto x.\mathrm{val} - 1$ (residues $1 \ldots p-1$ to indices $0 \ldots p-2$), inverse $k \mapsto (k+1)$. Both directions are checked; the range proof uses that a nonzero residue has value $0 < x.\mathrm{val} < p$, so x.val $- 1 < p - 1$. This makes the "circle of size $p - 1$" statement self-contained.

Fermat via Lagrange. A coprime a maps to the unit group via `unitOfCoprime`; the order of that unit divides the group order by Lagrange (`orderOf_dvd_card`), and `pow_orderOf_eq_one` gives $u^k = 1$ for $k = \mathrm{order}(u) \mid p - 1$. The congruence is recovered by casting back from $(\mathbb{Z}/p\mathbb{Z})^{\times}$ to $\mathbb{Z}/p\mathbb{Z}$ and from $\mathbb{Z}/p\mathbb{Z}$ to $\mathbb{N}$ via `natCast_eq_natCast_iff`.

Complete form by congruence lifting. The non-coprime case ($p \mid a$) is handled by the observation that $a \equiv 0 \pmod{p}$ (`Nat.mod_eq_zero_of_dvd`), lifted to powers by `Nat.ModEq.pow`: $a^p \equiv 0^p = 0 \pmod{p}$. Both sides congruent to 0 gives $a^p \equiv a$. This is the cleanest expression of "on the circle, $p \mid a$ means the point is the zero of the ring".

Verification. Compiled with Lean 4.32.2 (mathlib v4.32.2), zero errors, zero `sorry`, zero axioms beyond mathlib's. Build command in the artifact.

4 Relation to the pat exercise

The original insight (pat exercise XXIV) was formulated inside the pat framework. This re-formalization shows that the mathematical content survives the removal of the framework — and of the complex plane: the "prime axis as a circle" is exactly the modular circle $\mathbb{Z}/p\mathbb{Z}$ and its multiplicative group, entirely first-order algebra over a finite field. The pat vocabulary is preserved in the exercise record (README) as provenance, not in the proof.

5 Conclusion

The second 0pat exercise is complete: the prime axis as a circle — Fermat's little theorem, the size of the multiplicative circle group, and the infinitude of primes — proved with pure known mathematics in four theorems, zero errors. The observation requires no complex analysis: the circle structure of the primes modulo p is the algebra of $(\mathbb{Z}/p\mathbb{Z})^{\times}$. The method (genealogy-map path-walking) is validated on a second, algebraically different exercise.

References